INORGANIC CUMULATIVE EXAM

February 11, 1995

1) Describe the differences in the bonding for the methyl ligand in each of the following:

$Li_4(CH_3)_4$ $\qquad\qquad$ $CH_3Mn(CO)_5$

2) Briefly describe the bonding for the C_2H_4 ligand in $PtCl_3(C_2H_4)^{1-}$.

3) The ligand C_3H_5 can coordinate in two distinctly different modes.

   a) Draw partial structures to illustrate these two isomeric structures.

   b) Indicate how you would distinguish between these two possibilities spectroscopically.

   c) Give the formulas for two reasonable complexes of this ligand and the preparation for one of these complexes.

<u>(The compounds in #4 cannot be used for #3)</u>

4) Two compounds of the C_3H_5 ligand are shown below. Propose structures for both and indicate the driving force for the reaction. What are the most convenient reaction conditions for the transformation?

$$MN(C_3H_5)(CO)_5 \longrightarrow Mn(C_3H_5)(CO)_4 + CO$$

5) Several other common ligands show two or more different coordination modes. Give examples of five of these and indicate <u>all</u> of the commonly observed coordination modes for each of these ambidentate ligands.